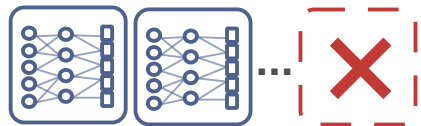


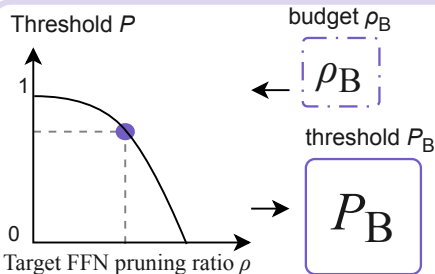
Offline calibration

Conventional: Full-or-Zero

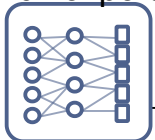


□ Full (high-fidelity expert) X Zero (not used) □ Light (low-fidelity expert)

Ours: Full-or-Light



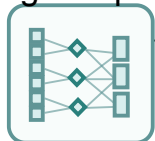
Full expert



Full output

Measure the gap

Light expert



Light output

Missing in Light

Token 1 gap

Token 2 gap

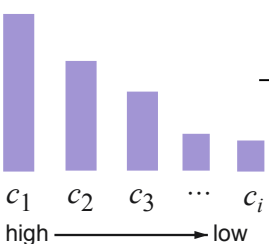
Token N gap

Stored Fidelity Gap

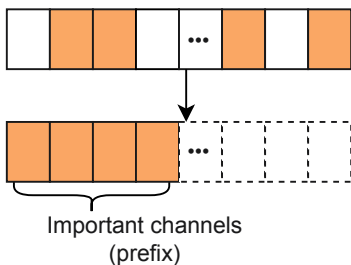
$\tilde{h}_i$

Used by **PAA** to decide which routed experts should be upgraded from Light to Full.

Channel score $r_{i,c}^{\text{act}}$



Reordered channels



0~1

prefix coverage for expert_i

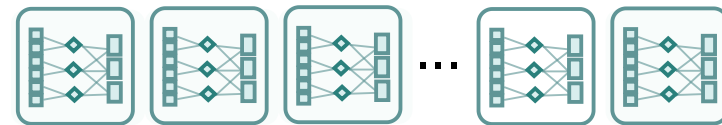
Stored prefix-coverage statistic

c_i

Used by **OMR** to estimate how much response is observed from Light experts.

Online inference

1 Top-K routed experts



2 PAA score

$$\hat{R}_i^{\text{PAA}}(x) = g_i(x) \times a_i^{\text{ACE}} \times \tilde{h}_i$$

3 Rank & select by cumulative mass

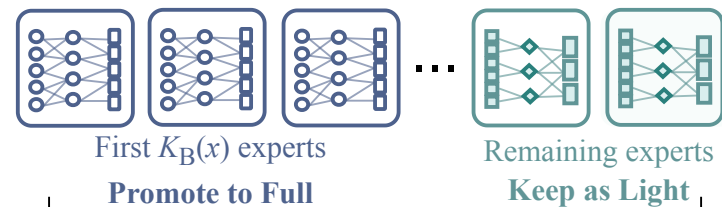
Routed experts sorted by $\hat{R}^{\text{PAA}}$	Promotion probabilities $p_{(j)}(x)$	Cumulative mass
$r_{(1)}$	$p_{(1)}(x)$	$p_{(1)}(x)$
$r_{(2)}$	$p_{(2)}(x)$	$p_{(1)}(x) + p_{(2)}(x)$
$r_{(3)}$	$p_{(3)}(x)$	$p_{(1)}(x) + p_{(2)}(x) + p_{(3)}(x)$
$\vdots$	$\vdots$	$\vdots$
$r_{(k)}$	$p_{(k)}(x)$	$\vdots$

Always retain the router top-1 expert

Select cutoff

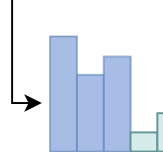
$$K_B(x) = \min \{ m : \sum_{j=1}^m p_{(j)}(x) \geq P_B \}$$

4 Promote top $K_B(x)$ to Full



5 OMR rescaling (using c_i)

Aggregation



$$m_O = \sum_{i \in \mathcal{F}} g_i + \sum_{i \in \mathcal{L}} g_i c_i$$

$$\hat{y} = \frac{y_O}{\max(m_O, \epsilon_m)}$$

